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JAVA PROJECT REPORT ON

Password Vault and Credential Storage System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**Password Vault and Credential Storage System"** submitted in partial fulfillment of the requirements for the course Java Programming is a **record of my original work** carried out under the guidance of **MR.ARAVIND** , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Shree Nisha.R

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Password Vault and Credential Storage System"** is a **bona fide work** carried out by Shree Nisha.R**, 1 - 'c'/2025**, BSC Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on___10/04/2026_____

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The Password Vault and Credential Storage System is a Java-based application designed to securely store and manage user credentials such as usernames and passwords. In today's digital world, users maintain multiple accounts across various platforms, making it difficult to remember and manage passwords efficiently. Storing passwords manually or using insecure methods can lead to security risks and data loss.This project aims to provide a simple and organized system that allows users to store, view, and manage their credentials in a structured manner. The system enables users to add account details such as website name, username, and password, and retrieve them when needed. It helps reduce the difficulty of remembering multiple passwords and improves overall data management.The application is developed using Core Java programming concepts, including classes, objects, methods, loops, conditional statements, and collections such as ArrayList. The system is implemented as a menu-driven console application, making it simple and easy to use.The main objective of this system is to improve password management, enhance organization of credentials, and reduce the risk of forgetting important login details. This project also demonstrates the practical implementation of object-oriented programming concepts in Java and provides a foundation for developing more advanced secure password management systems in the future.

1. INTRODUCTION

In today's digital era, individuals use multiple online platforms such as social media, banking applications, email services, and various websites that require secure login credentials. Managing and remembering different usernames and passwords for each account can be difficult and often leads to poor practices such as reusing passwords or writing them down in insecure places. This increases the risk of unauthorized access and data breaches.To overcome these challenges, there is a need for a system that can securely store and manage user credentials in an organized manner. A Password Vault and Credential Storage System provides a centralized solution where users can store their login details and retrieve them whenever required. This helps in improving password management and reduces the chances of forgetting important credentials.The Password Vault and Credential Storage System is a Java-based application developed to provide a simple and efficient way of managing user credentials. The system allows users to store account details such as website name, username, and password, and view or delete them when necessary. It follows a menu-driven approach, making it easy for users to interact with the system.This project is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and collections like ArrayList. The system demonstrates how Java can be used to build practical applications for real-world problems.The primary goal of this project is to simplify credential management, improve data organization, and provide a basic foundation for developing more advanced and secure password management systems in the future.

1.1 Objective

The primary objective of the Password Vault and Credential Storage System is to develop a simple, efficient, and user-friendly application using Core Java to store and manage user credentials securely.

The specific objectives of this project are:

- To design and develop a Java-based system for storing user credentials such as website name, username, and password
- To provide an easy method for users to manage multiple account details in one place
- To reduce the difficulty of remembering multiple passwords
- To improve organization and accessibility of stored credentials
- To demonstrate the use of Core Java concepts such as classes, objects, methods, loops, and collections
- To provide a simple and menu-driven interface for easy user interaction
- To create a basic foundation for future enhancements such as encryption and database integration

1.2 Overview

The Password Vault and Credential Storage System is a menu-driven application developed using the Java programming language. The system provides a simple and organized platform for users to store and manage their login credentials in one place.The application allows users to add details such as website or application name, username, and password. These credentials are stored in the system and can be viewed or deleted whenever required. This helps users manage multiple accounts efficiently without the need to remember each password manually.The system uses Java collections such as ArrayList to store credential data dynamically during program execution. The application follows a structured approach where each functionality, such as adding, viewing, and deleting credentials, is implemented as separate methods.The system is designed to be simple, user-friendly, and easy to operate through a console-based interface. It is suitable for beginners learning Java programming as well as for small-scale personal use.This project demonstrates how Java programming can be used to develop real-world applications that solve common problems like password management and data organization.

1.3 Features

The Password Vault and Credential Storage System includes the following key features:

- **Credential Storage** – Allows users to store account details such as website or application name, username, and password
- **Add Credentials** – Enables users to add new login details into the system easily
- **View Credentials** – Displays all stored credentials in a clear and organized format
- **Delete Credentials** – Allows users to remove unwanted or outdated credentials from the system
- **User-Friendly Interface** – Provides a simple menu-driven interface for easy interaction
- **Fast Access to Data** – Users can quickly retrieve stored credentials whenever required
- **Dynamic Data Handling** – Uses ArrayList to store data dynamically during program execution
- **Error Reduction** – Minimizes the chances of forgetting or misplacing passwords

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should function and what features are required to solve the problem efficiently.In the context of the Password Vault and Credential Storage System, program analysis focuses on identifying the challenges faced by users in managing multiple passwords and providing a simple and organized solution using Java programming.

2.1 Problem Definition

In today's digital environment, users have multiple accounts across different platforms such as social media, banking, email, and online services. Managing login credentials for all these accounts becomes difficult.

Some common problems include:

- Difficulty in remembering multiple passwords
- Use of weak or repeated passwords
- Storing passwords in insecure places such as notebooks or plain text files
- Risk of unauthorized access and data loss
- Lack of proper organization of credentials

These issues highlight the need for a secure and efficient system to store and manage user credentials. The Password Vault System is designed to solve these problems by providing a structured and easy-to-use application.

2.2 Project Overview

The Password Vault and Credential Storage System is a Java-based application developed to help users manage their login credentials effectively.The system provides a menu-driven interface where users can perform operations such as adding new credentials, viewing stored credentials, and deleting existing credentials.

The working of the system is as follows:

- The user starts the application
- A menu with options is displayed
- The user selects an option based on the required operation
- The system processes the input and performs the action
- The result is displayed to the user
- The process continues until the user exits

Technology Used

- Programming Language: Java
- Concepts Used: OOP (Classes, Objects), ArrayList, Loops, Conditional Statements

Advantages of the System

- Easy to use
- Helps manage multiple credentials
- Reduces the need to remember passwords
- Organized data storage

2.3 Scopes and Limitations

Scope of the System

The scope of the Password Vault System includes:

- **Storing user credentials securely in a structured format**
- **Managing multiple account details in one application**
- **Allowing users to add, view, and delete credentials**
- **Providing a simple interface for easy usage**
- **Demonstrating practical implementation of Java programming concepts**

Applications of the System

- **Personal password management**
- **Educational purposes**
- **Small-scale applications**

Limitations of the System

Despite its advantages, the system has some limitations:

- **No encryption for stored passwords (basic version)**
- **No database integration (data not stored permanently)**
- **Console-based interface (no graphical interface)**
- **Single-user system**

Future Enhancements

The system can be improved by:

- **Adding password encryption for better security**
- **Integrating a database for permanent storage**
- **Developing a graphical user interface (GUI) using Java Swing or JavaFX**
- **Adding authentication (login system)**
- **Implementing advanced security features**

3. SYSTEM DESIGN

System design is the phase where the structure and components of the system are planned before implementation. It defines how the system will operate, how different modules interact, and how data is processed. A well-designed system improves efficiency, maintainability, and scalability.The Password Vault and Credential Storage System is designed as a menu-driven console application using Java. The system follows a modular approach, where each module performs a specific function such as storing, managing, and displaying user credentials.The design ensures simplicity, ease of use, and efficient handling of credential data.

3.1 Modules

The system is divided into the following modules:

---

1. Credential Management Module

Description:
This module is responsible for handling all operations related to user credentials.

Functions:

- Add new credentials
- Store website name, username, and password
- Display stored credentials

Working:
When the user selects the option to add credentials, the system accepts input details and stores them in an ArrayList.

Importance:
This is the core module of the system, as it manages all credential-related data.

2. Storage Module

Description:
This module handles the storage of credential data during program execution.

Functions:

- Store credentials dynamically
- Maintain list of all entries

Working:
The system uses ArrayList to store credential objects. Each object contains site name, username, and password.

Importance:
Ensures efficient and flexible data storage without fixed size limitations.

3. Retrieval Module

Description:
This module is responsible for retrieving and displaying stored credentials.

Functions:

- View all stored credentials
- Display details in organized format

Working:
When the user selects the view option, the system retrieves all stored data from ArrayList and displays it.

Importance:
Allows quick and easy access to stored information.

4.Deletion Module

Description:
This module allows users to remove unwanted or outdated credentials.

Functions:

- Search credentials by website name
- Delete selected credential

Working:
The system compares user input with stored data and removes the matching entry.

Importance:
Helps maintain updated and clean data storage.

5. User Interface Module

Description:
This module manages user interaction with the system.

Functions:

- Display menu options
- Accept user input
- Navigate between operations

Working:
A menu-driven interface is displayed, and the user selects options to perform operations.

Importance:
Ensures the system is easy to use and understand.

3.2 System Architecture

The system follows a simple architecture:

- **User Interface (Console Input/Output)**
- **Processing Logic (Java Classes and Methods)**
- **Data Storage (ArrayList in Memory)**

Working Flow:

- User enters input through console
- System processes input using Java logic
- Data is stored in ArrayList
- Output is displayed back to the user

3.3 Data Design

The system stores data in the form of objects.

Credential Data:

- Website/Application Name
- Username
- Password

All data is stored using ArrayList, which allows dynamic storage during runtime.

3.4 Input Design

Input design focuses on how data is entered into the system.

Types of Input:

- Website name
- Username
- Password

Validation:

- Ensure valid input is entered
- Avoid empty or incorrect values

3.5 Output Design

Output design defines how results are displayed.

Types of Output:

- Display stored credentials
- Confirmation messages (added/deleted)

Characteristics:

- Clear and readable format
- Proper labeling
- Organized structure

3.6 System Workflow

The workflow of the system is as follows:

Start
↓
Display Menu
↓
User Selects Option
↓
Perform Operation (Add / View / Delete)
↓
Display Result
↓
Repeat or Exit
↓
End

3.7 Advantages of System Design

- Simple and easy to understand
- Modular structure
- Efficient data handling
- Easy to maintain and update

3.8 Limitations of Design

- No permanent storage (no database)
- No encryption for passwords
- No graphical interface

4. METHODOLOGY

Methodology refers to the systematic approach followed to develop the project. It describes the sequence of steps taken from understanding the problem to designing, implementing, testing, and documenting the system.For the Password Vault and Credential Storage System, a structured development approach was followed to ensure that the system meets user requirements and functions efficiently. Each phase was carefully planned and executed to achieve accurate and reliable results.

The methodology used in this project includes the following phases:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

Each phase plays an important role in the successful development of the system.

4.1. Problem Analysis

Problem analysis is the first step in the development process. In this phase, the existing issues related to password management are studied and analyzed.

Activities Performed:

- Identified problems in managing multiple passwords

- Studied insecure methods of storing credentials
- Defined system requirements

Key Observations:

- Users forget passwords frequently
- Passwords are stored in insecure places
- Difficult to manage multiple accounts

Outcome:

A clear understanding of the need for a secure and organized credential management system was established.

4.2. 4.2 System Design

In this phase, the overall structure of the system is planned before coding begins.

Activities Performed:

- Divided the system into modules
- Designed workflow of the application
- Defined data structure (Credential class)
- Planned input and output formats

Design Approach:

- Modular design
- Menu-driven interface

- Object-Oriented Programming (OOP)

Outcome:

A well-structured system design that serves as a blueprint for implementation.

4.3. Program Development

This phase involves writing the actual program using Java.

Activities Performed:

- Created Credential class
- Implemented menu-driven logic
- Developed methods for add, view, and delete operations
- Used ArrayList for storing data
- Applied loops and conditional statements

Concepts Used:

- Classes and Objects
- Methods
- Loops (for, while)
- Conditional statements (if-else, switch)
- Collections (ArrayList)

Outcome:

A fully functional program capable of managing user credentials.

4.4. Testing and Validation

Testing ensures that the system works correctly and produces accurate results.

Types of Testing Performed:

1. **Functional Testing**
   Checked whether all features work correctly

2. **Input Testing**
   Tested with valid and invalid inputs

3. **Output Testing**
   Verified correctness of displayed results

4. **Error Testing**
   Checked handling of incorrect or missing inputs

Test Case	Input	Expected Output
Add Credential	Site, Username	Credential stored
View Data	—	Display all credentials
Delete Data	Site name	Credential removed
Invalid Input	Empty/Incorrect	Error handled

Outcome:

The system performs all operations correctly and handles different scenarios effectively.

4.5. Result and Documentation

After successful testing, the results of the system are documented.

Activities Performed:

- Verified program outputs
- Recorded sample inputs and outputs
- Prepared project report
- Documented system design and working

Outcome:

A complete and well-documented project ready for submission.

4.6. Algorithm

The algorithm describes the step-by-step procedure of the system.

Algorithm Steps:

1. Start the program
2. Display menu options
3. Accept user choice
4. Perform selected operation
    - Add Credential
    - View Credentials
    - Delete Credential
5. Display result
6. Repeat until exit option is selected
7. End the program

4.7. Flowchart

A flowchart represents the workflow of the system.

Flow Description:

Start
↓
Display Menu
↓
Enter Choice
↓
Perform Operation
↓
Display Result
↓
Repeat / Exit
↓
End

Tools for Drawing Flowchart:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

This section contains the important parts of the Java program used to develop the Password Vault and Credential Storage System. The program is implemented using Core Java concepts such as classes, objects, methods, loops, and conditional statements.The system uses an **ArrayList** to store credential data dynamically during program execution. Each credential is stored as an object containing the website name, username, and password.

Main Class Structure

The main class contains the menu-driven logic and controls the overall execution of the program.

```
public class Main {
    static ArrayList<Credential> vault = new ArrayList<>();
}
```

Credential Class

The Credential class is used to store user data.

```
class Credential {
    String site;
    String username;
    String password;

    Credential(String site, String username, String password) {
        this.site = site;
        this.username = username;
        this.password = password;
    }
}
```

Adding Credential

This part of the code allows the user to add new credentials.

```
static void addCredential() {
    System.out.print("Enter Website: ");
    String site = sc.nextLine();

    System.out.print("Enter Username: ");
    String username = sc.nextLine();

    System.out.print("Enter Password: ");
    String password = sc.nextLine();

    vault.add(new Credential(site, username, password));
    System.out.println("Credential Saved!");
}
```

Viewing Credentials

This code displays all stored credentials.

```
static void viewCredentials() {
    for (Credential c : vault) {
        System.out.println("Site: " + c.site +
                           " Username: " + c.username +
                           " Password: " + c.password);
    }
}
```

Deleting Credential

This code removes a credential based on website name.

```
static void deleteCredential() {
```

```
    System.out.print("Enter Website to delete: ");
    String site = sc.nextLine();

    Iterator<Credential> it = vault.iterator();

    while (it.hasNext()) {
        Credential c = it.next();
        if (c.site.equalsIgnoreCase(site)) {
            it.remove();
            System.out.println("Deleted Successfully!");
            return;
        }
    }

    System.out.println("Not Found!");
}
```

Menu Logic

This part controls the user interaction.

```
do {
    System.out.println("1. Add");
    System.out.println("2. View");
    System.out.println("3. Delete");
    System.out.println("4. Exit");

    choice = sc.nextInt();
    sc.nextLine();

    switch (choice) {
```

```
        case 1: addCredential(); break;
        case 2: viewCredentials(); break;
        case 3: deleteCredential(); break;
    }

} while (choice != 4);
```

Explanation of Code

- The program uses a **menu-driven approach** for user interaction
- **ArrayList** is used to store credential objects dynamically
- **Classes and Objects** are used to organize data
- **Loops and Conditions** are used to control program flow
- Each function performs a specific operation such as add, view, and delete

6. INPUT AND OUTPUT SCREENS

This section shows the sample inputs and outputs of the Password Vault and Credential Storage System. It demonstrates how the system interacts with the user and displays results.Students can also include screenshots of program execution to support these outputs.

Input Example

===== PASSWORD VAULT SYSTEM =====

1. Add Credential

2. View Credentials

3. Delete Credential

4. Exit

Enter choice: 1

Enter Website/App Name: Gmail

Enter Username: user123

Enter Password: pass123

Output Example

Credential Saved!

View Credentials Output

--- Stored Credentials ---

Site: Gmail | Username: user123 | Password: pass123

Delete Credential Input

Enter Website/App Name to delete: Gmail

Delete Credential Output

Credential Deleted!

Invalid Case Output

Not Found!

Sample Menu Screen

===== PASSWORD VAULT SYSTEM =====

1. Add Credential

2. View Credentials

3. Delete Credential

4. Exit

Screenshots (To be Added)

You can add screenshots with labels like:

- **Figure 1:** Menu Screen
- **Figure 2:** Add Credential Screen
- **Figure 3:** View Credentials Output
- **Figure 4:** Delete Credential Output

```
===== PASSWORD VAULT SYSTEM =====
1. Add Credential
2. View Credentials
3. Delete Credential
4. Exit
Enter your choice:
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Password Vault and Credential Storage System was developed to provide a simple and efficient solution for managing user credentials. The system allows users to store, view, and delete login details such as website name, username, and password in an organized manner.During the development of this project, Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and collections like ArrayList were effectively used. These concepts helped in creating a structured and modular application where each functionality is handled separately.The system was tested with different input scenarios to ensure proper functioning. It was observed that the application successfully stores credentials, retrieves them when required, and deletes them efficiently. The menu-driven interface makes the system easy to use, even for beginners.One of the main advantages of this system is that it helps users manage multiple credentials without confusion and reduces the chances of forgetting important passwords. However, the system has some limitations, such as lack of encryption and absence of permanent data storage.Overall, the project demonstrates how Java programming can be used to develop simple real-world applications and provides a strong foundation for further enhancements.

7.2 Conclusion

The Password Vault and Credential Storage System successfully demonstrates the practical implementation of Java programming in developing a useful application for managing credentials. The system provides a simple, user-friendly interface to store and manage login details efficiently.This project highlights the importance of organized password management and shows how basic programming concepts can be applied to solve real-life problems. The application reduces manual effort, improves accessibility of stored data, and enhances overall efficiency.Although the current system is basic, it can be further improved by adding advanced features such as password encryption, database integration for permanent storage, graphical user interface using Java Swing or JavaFX, and authentication mechanisms for better security.In conclusion, this project serves as a strong foundation for developing more secure and advanced password management systems and demonstrates the effective use of Java in real-world applications.

8. REFERENCES

The following resources were used during the development of this project:

Books

Books

- *Herbert Schildt – Java: The Complete Reference*
- *E. Balagurusamy – Programming with Java*

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